

Employee Compensation and State Pension Performance

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- The pension's board management offer a linear contract:
 $salary = base + bonus * r * liab * AUM$ to the investment executive at $t = 0$
- where $base$ is the base salary paid to the investment executive and $bonus$ is a rate tied to his output return (r). The liability rate ($liab$) and assets under management (AUM) are exogenously given
- $bonus \in [0, 1]$ and is rate on the output paid to the investment executive as bonus
- Lets assume $r \in [-\infty, \infty]$ is the rate of return actually observed by the pension board
- Let the investment executive choose a costly risk level β in relation to the liability rate $\beta \in [-\infty, \infty]$ for the portfolio. The output is observed at $t=1$
- The output observed by the pension holders is assumed to be:

$$r * liab * AUM = \beta * liab * AUM + \epsilon * liab * AUM \quad (1)$$

- where $\epsilon \sim \mathcal{N}(0, \sigma^2)$
- Management board is risk neutral and profit function is as follows

$$\mathbb{E}[r * liab * AUM - salary(r)] \quad (2)$$

- The investment executive's is assumed to be risk averse to have the following CARA utility

$$U(salary, \beta) = \mathbb{E}[1 - e^{-\lambda(salary(r) - c * liab * AUM)}] \quad (3)$$

c is the cost of implementing β which we assume to follow a quadratic cost: $c(\beta) = c * \frac{\beta^2}{2}$

- λ is the risk aversion coefficient of the investment executive

The investment executive is a utility maximizer after observing the contract and the board management maximizes potential profit, however, the risk level chosen by the investment executive is hidden to management.

1 First Best

Lets assume the agent's risk level is contractible by mangagement and is not a hidden action

$$\begin{aligned} \max_{\beta} \quad & \mathbb{E}[\beta * liab * AUM + \epsilon * liab * AUM - base - \\ & bonus * r * liab * AUM] \\ s.t. \quad & \mathbb{E}[1 - e^{-\lambda(salary(r) - c * liab * AUM)}] \geq U_0 \text{ (IR Constraint)} \end{aligned} \quad (4)$$

Management choice is maximizing. Thus the optimal salary is independent of risk level β chosen and we can assume salary = base

we can rewrite the individual rationality constraint as below

$$\mathbb{E}[1 - e^{-\lambda(base - c(\beta) * liab * AUM)}] \geq U_0 \quad (5)$$

$$1 - e^{-\lambda(base - (.5 * c * (\beta)^2) * liab * AUM)} \geq U_0 \quad (6)$$

we can rewrite the managment's problem as below

$$\max \mathbb{E}[\beta * liab * AUM + \epsilon * liab * AUM - base] \quad (7)$$

$$\max[\beta * liab * AUM - base] \quad (8)$$

we now need to substitute the beta from the invidual rationality constraint in our maximizing equation

let $z = (.5 * c * (\beta)^2) * liab * AUM$

$$1 - e^{-\lambda(base - z)} = U_0 \quad (9)$$

$$base = -\frac{1}{\lambda} \ln(-U_0 + 1) + z \quad (10)$$

make the substitution into the objective function

$$\max \beta * liab * AUM + \frac{1}{\lambda} \ln(-U_0 + 1) - z \quad (11)$$

Apply FOC w/ respect to β

$$liab * AUM - \frac{\partial z}{\partial \beta} = 0 \quad (12)$$

$$\frac{\partial z}{\partial \beta} = liab * AUM \quad (13)$$

$$(c * (\beta)) * liab * AUM = liab * AUM \quad (14)$$

with simple algebra the optimal beta is the following:

$$\beta^* = \frac{1}{c} \quad (15)$$

The risk level taken is inversely related with the costs of implementing beta

Let us consider the moral hazard where managment cannot force the investment executive to chose optimal risk level β

First we maximize the general case with respect to investment executive for an arbitrary risk level β

$$\max_{\beta} \mathbb{E}[1 - e^{-\lambda[base+bonus*(\beta*liab*AUM+\epsilon*liab*AUM)-z]}] \quad (16)$$

$$\max_{\beta} [1 - e^{-\lambda[base+bonus*\beta*liab*AUM-z]}] \mathbb{E}[e^{-\lambda*bonus*liab*AUM*\epsilon}] \quad (17)$$

$$\max_{\beta} [1 - e^{-\lambda(base+bonus*\beta*liab*AUM-z)}][e^{.5*\lambda^2*bonus^2*liab^2*AUM^2*\sigma^2}] \quad (18)$$

$$\max_{\beta} [1 - e^{-\lambda(base+bonus*\beta*liab*AUM-z-.5*\lambda*bonus^2*liab^2*AUM^2*\sigma^2)}] \quad (19)$$

The above is the same as maximizing the below

$$\max_{\beta} [base+bonus*\beta*liab*AUM-z] - .5*\lambda*bonus^2*liab^2*AUM^2*\sigma^2] \quad (20)$$

Apply the FOC with respect to β

$$bonus * liab * AUM - c * \beta * liab * AUM = 0 \quad (21)$$

$$bonus - c * \beta = 0 \quad (22)$$

$$\beta^* = \frac{bonus}{c} \quad (23)$$

The bonus rate can never be greater than 1, thus investment executive takes a risk level that is sub-optimal to the first best solution.

The principal knowing this chooses to maximize

$$\begin{aligned} \max \quad & \mathbb{E}[\beta * liab * AUM + \epsilon * liab * AUM - base - bonus(\beta * liab * AUM + \epsilon)] \\ \text{s.t.} \quad & \beta = bonus/c \\ & [1 - e^{-\lambda(base + \beta * bonus * liab * AUM - z - .5 * \lambda * bonus^2 * liab^2 * AUM^2 * \sigma^2)}] \geq U_0 \end{aligned} \quad (24)$$

We can rewrite the objective function:

$$\max \quad bonus/c * liab * AUM * (1 - bonus) - base \quad (25)$$

Solve the incentive rationality with plugging in the incentive compatibility constraint

$$base + liab * AUM * bonus^2 / c - z - .5 * \lambda * bonus^2 * liab^2 * AUM^2 * \sigma^2 = \ln(U_0 - 1) / \lambda \quad (26)$$

$$base = \ln(U_0 - 1) / \lambda - (liab * AUM * bonus^2 / c - z - .5 * \lambda * bonus^2 * liab^2 * AUM^2 * \sigma^2) \quad (27)$$

sub the IR constraint back into the object function maximization

$$\begin{aligned} \max \quad & bonus/c * liab * AUM * (1 - bonus) - \ln(U_0 - 1) / \lambda + \\ & (liab * AUM * bonus^2 / c - z - .5 * \lambda * bonus^2 * liab^2 * AUM^2 * \sigma^2) \end{aligned} \quad (28)$$

Note that $z = \frac{liab * AUM * bonus^2}{2c}$

Take FOC with respect to *bonus*

$$1/c * liab * AUM - 2 * bonus/c * liab * AUM + 2 * bonus/c * liab * AUM - liab * AUM * bonus/c - \lambda * bonus liab^2 * AUM^2 * \sigma^2 = 0 \quad (29)$$

$$1/c * liab * AUM - liab * AUM * bonus/c - \lambda * bonus * liab^2 * AUM^2 * \sigma^2 = 0 \quad (30)$$

$$1/c - bonus/c - \lambda * bonus * liab * AUM * \sigma^2 = 0 \quad (31)$$

$$bonus/c + \lambda * bonus * liab * AUM * \sigma^2 = 1/c \quad (32)$$

multiply through by c

$$bonus(1/c + \lambda * liab * AUM * \sigma^2) = 1/c \quad (33)$$

$$bonus^* = \frac{1}{1 + c * \lambda * liab * AUM * \sigma^2} \quad (34)$$

The bonus is lower powered as the costs of implementing β increases, risk aversion increases, size of the fund increases and σ increases.

2 Multi-Banit Problem

Let's assume an infinite horizon Markov decision problem in discrete time.

Let time be indexed by $t = 0, 1, 2, 3, \dots$

At each t , the CEO chooses amongst r return levels on investment and we denote this choice by actions $a_t \in 1, \dots, R$

Let $a_t = r$, a stochastic return is realized x_t^r which is simply $r + \epsilon$. The agent is seeking to maximize his utility, which at each point in time is $-e^{-\lambda(\text{salary} + \text{bonus} * \max(x_t * \text{liab} * AUM, 0) - c)}$. Where c is his money equivalent of personal cost. The bonus rate is given by the board and is endogenously defined by an independent bandit process.

At each point in time, the CEO learns more about the distribution of r and maximizes utility. We assume a complete market where every possible r can be chosen. Let the discovery process about the utility of arm r be represented by a recursive exponential moving process with a learning rate of α .

$$u_t(r) = \begin{cases} u_t(x_t^r) & \text{if } n = 1 \\ u_t(x_t^r)(1 - \alpha) + \alpha u_{t-1}(r) & \text{if } n \geq 1 \end{cases} \quad (35)$$

where n is the number of times r has been played. As α increases the speed of learning increases. A recursive moving average can also be considered as well as alternative weighting schemes. r_0 will be assumed to be 1.0.

We assume the CEO is utility-maximizing and chooses r with the highest utility $u_t(r)$ is a pure exploitation strategy. To force exploration, we define an ϵ -Greedy search process where we explore other choices of r with probability ϵ . Programatically this is found by just drawing from the uniform

interval and playing a random r if a number is drawn less than or equal to ϵ .

Simultaneously we assume the board chooses amongst b bonus rate levels on output and we denote this choice by action $z_t \in 1, \dots, B$

Let $z_t = b$ for a single iteration. The bonus rate unlike the risk return level of the CEO is non stochastic. The board is utility maximizing with their utility being defined as simple output $return * Liab * AUM - base - bonus$. I is exogenously given and the return is endogenously defined and given by the action of the CEO.

$$u_t(b) = \begin{cases} u_t(z_t^b) & \text{if } n = 1 \\ u_t(z_t^b)(1 - \alpha) + \alpha u_{t-1}(b) & \text{if } n \geq 2 \end{cases} \quad (36)$$

We let the board search space of actions by the same epsilon-greedy process as defined as above.

The choice of r and b are independent, and the state transition function only depends on the choice and the realized payoff. The corresponding payoff of the action is simply:

$$r_{greedy} = \operatorname{argmax}_a u_t(r) \quad (37)$$

$$b_{greedy} = \operatorname{argmax}_z u_t(b) \quad (38)$$

As t increases, the optimal choice of r and b will be found as additional exploration does not represent an information gain to the distribution of both $u_t(r)$ and $u_t(b)$.